

REPORT

Mutations in the Alpha 1,2-Mannosidase Gene, *MAN1B1*, Cause Autosomal-Recessive Intellectual Disability

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We have used genome-wide genotyping to identify an overlapping homozygosity-by-descent locus on chromosome 9q34.3 (MRT15) in four consanguineous families affected by nonsyndromic autosomal-recessive intellectual disability (NS-ARID) and one in which the patients show additional clinical features. Four of the families are from Pakistan, and one is from Iran. Using a combination of next-generation sequencing and Sanger sequencing, we have identified mutations in the gene *MAN1B1*, encoding a mannosyl oligosaccharide, alpha 1,2-mannosidase. In one Pakistani family, MR43, a homozygous nonsense mutation (RefSeq number NM_016219.3: c.1418G>A [p.Trp473*]), segregated with intellectual disability and additional dysmorphic features. We also identified the missense mutation c. 1189G>A (p.Glu397Lys; RefSeq number NM_016219.3), which segregates with NS-ARID in three families who come from the same village and probably have shared inheritance. In the Iranian family, the missense mutation c.1000C>T (p.Arg334Cys; RefSeq number NM_016219.3) also segregates with NS-ARID. Both missense mutations are at amino acid residues that are conserved across the animal kingdom, and they either reduce k_{cat} by ~1300-fold or disrupt stable protein expression in mammalian cells. *MAN1B1* is one of the few NS-ARID genes with an elevated mutation frequency in patients with NS-ARID from different populations.

Intellectual disability (ID), also called mental retardation (MR), is a devastating neurodevelopmental disorder that has a serious impact on the affected individuals and their families, as well as on health and social services. It is believed to occur with a prevalence of ~1%–3% within the population,^{1,2} and is frequently the result of genetic aberrations. ID can present as the sole clinical feature (nonsyndromic [NS]; see MIM 249500), or it can be present with additional clinical or dysmorphological features (syndromic [S]). Generally speaking, it is much more straightforward to identify the genetic cause of S-ID patients because the comorbid features frequently enable a medical geneticist to narrow down the suspected cause to a mutation in a short list of genes, or perhaps in a single gene. However, for individuals with NS-ID, no secondary clues assist the molecular diagnosis. ID is significantly more frequent in males than in females, and it had been assumed that ~25% of severe cases were X-linked; however, a recent review suggests that X-linked mutations contribute to no more than 10% of cases,³ and thus it is expected that there will be many more autosomal genes

whose mutations cause NS-ID, both dominant and recessive. However, as a result of the high degree of genetic heterogeneity, mutations in only seven genes have been reported to cause NS-ARID, and all of these have been identified on the basis of mapping regions of autozygosity or homozygosity-by-descent (HBD) in multiplex consanguineous families (reviewed in Kaufman et al.⁴).

We ascertained three large NS-ARID-affected multiplex consanguineous families from a farming community in the Dera Ghazi Khan district, within Punjab province in Pakistan (MR7, MR8, MR9), one from Sukkur in Sindh province (MR43), and one from Iran (8600060; Figure 1). Appropriate informed consent was obtained for all participants in the study, and institutional research ethics approval was obtained.

The description of ascertainment, clinical features, and mapping for families MR7, MR8, and MR9 are given elsewhere⁵ but are summarized in Tables 1 and 2. Two individuals from MR7 (IV:5 and IV:6, age 14 and 13 years, respectively) were assessed and found to have similar developmental history. Both had delayed developmental

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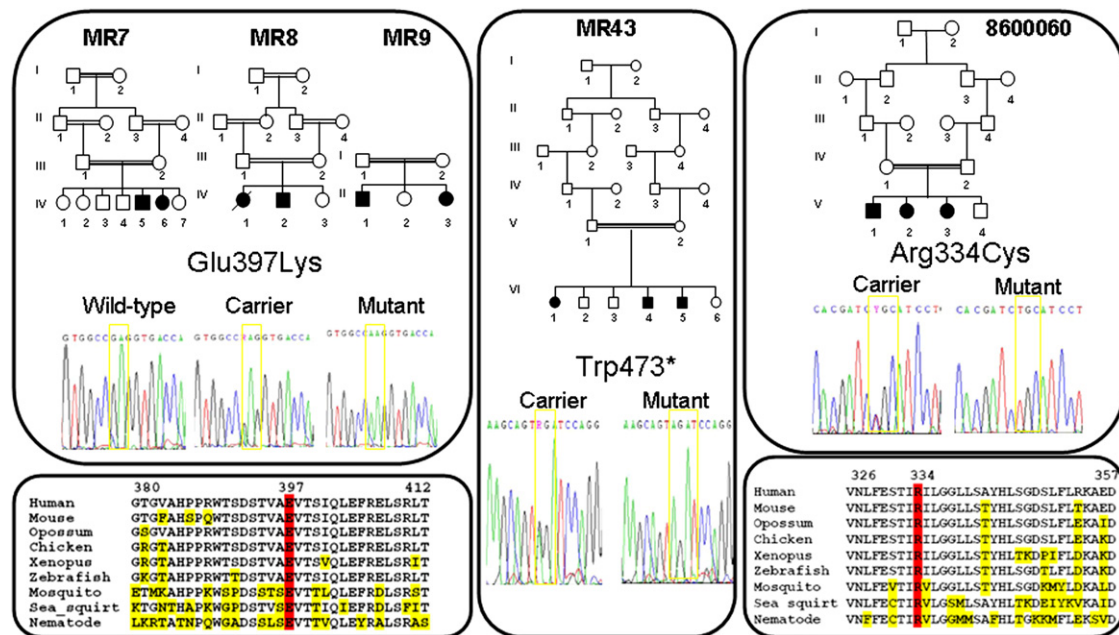


Figure 1. Pedigrees of Families MR7, MR8, and MR9 from Dera Ghazi Khan, and from MR43 from Sukkur in Pakistan, and 8600060 from Iran

Filled circles indicate affected individuals. Birth order is not assigned for generation VI in family MR43. Below the pedigrees, electropherograms from sequence analysis of *MAN1B1* are shown, indicating the p.Glu397Lys, p.Trp473*, and p.Arg334Cys missense mutations, respectively. For the p.Glu397Lys and p.Arg334Cys missense mutations, the panel below the electropherograms shows CLUSTAL 2.1 multiple sequence alignment indicating conservation of the substituted amino acids in *MAN1B1* orthologs across the animal kingdom in red. Nonconserved residues are shown with yellow highlighting. Photographs of the affected individuals are available on request from the corresponding author (signing of a confidentiality and nondisclosure agreement is required).

milestones and began walking at age 4 yr. Speech was also delayed, but they are now both able to speak in sentences, albeit limited to immediate and day-to-day needs. They are toilet trained but require assistance with bathing and dressing. Both have hyperphagia and are overweight. MR7

IV:5 also suffers from asthma. Brain MRI scan for MR7-IV:5 was examined by a radiologist (GI), and showed no evidence of white-matter abnormality within the cerebellum or cerebrum; however, we observed a small prominent perivascular space in the right parietal lobe, as well as

Table 1. Summary of Clinical Information

Family ID	Mutation (Nucleotide; Protein)	Gender	Age	IQ	Syndromic/ Nonsyndromic	OFC (cm)	Height (cm)	Obesity (Weight in Kg)	Epilepsy
MR7- IV:5	c.1189G>A; p.Glu397Lys	M	14	45–55	NS	55.5	155	+ (68)	-
MR7- IV:6	c.1189G>A; p.Glu397Lys	F	13	45–55	NS	54.5	150	+ (60)	-
MR8- IV:2	c.1189G>A; p.Glu397Lys	M	10	50–60	NS	NK ^a	NK ^a	-	-
MR8- IV:1 ^b	c.1189G>A; p.Glu397Lys	F		N.K.	NS	NK ^a	NK ^a	-	-
MR9-II-3	c.1189G>A; p.Glu397Lys	F	9	45–55	NS	51	119	- (23)	-
MR9-II-1	c.1189G>A; p.Glu397Lys	M	18	45–55	NS	NK ^a	NK ^a	-	+
MR43-VI:4	c.1418G>A; p.Trp473*	M	9	55–65	S	53	130	- (26.9)	+
MR43-VI:1	c.1418G>A; p.Trp473*	F	16	55–65	NK ^a	55	138	- (36)	-
MR43-VI:5	c.1418G>A; p.Trp473*	M	8	55–65	S	53	121	- (19)	-
8600060-V:1	c.1000C>T; p.Arg334Cys	M	11	59	NS	53	136	-	-
8600060-V:2	c.1000C>T; p.Arg334Cys	F	25	63	NS	55	155	-	-
8600060-V:3	c.1000C>T; p.Arg334Cys	F	23	56	NS	53.5	154	-	-

Nucleotide sequence uses NM_016219.3, the reference mRNA sequence for *MAN1B1*.

^a NK: not known or not measured.

^b Now deceased (from hepatitis C).

Table 2. Details of Examination of Photographs of Affected Family Members

Family ID	Mutation (Nucleotide; Protein)	Physical Features and Dysmorphology				
		Eyes	Head and Ears	Face	Hands	Feet
MR7- IV:5	c.1189G>A; p.Glu397Lys	down-slanting palpebral fissures; thin, sparse eyebrows	mild dolicocephaly; short neck	broad nose		
MR7- IV:6	c.1189G>A; p.Glu397Lys	down-slanting palpebral fissures; thin, sparse eyebrows	mild dolicocephaly; short neck	broad nose		
MR8- IV:2	c.1189G>A; p.Glu397Lys	NK ^a	NK ^a	NK ^a	NK ^a	NK ^a
MR8- IV:1 ^b	c.1189G>A; p.Glu397Lys	NK ^a	NK ^a	NK ^a	NK ^a	NK ^a
MR9-II:3	c.1189G>A; p.Glu397Lys	Arched, sparse eyebrows;		triangular, pointed chin		
MR9-II:1	c.1189G>A; p.Glu397Lys	Arched, sparse eyebrows; long eyelashes		triangular, pointed chin flat philtrum	NK ^a	NK ^a
MR43-VI:4	c.1418G>A; p.Trp473*	down-slanting palpebral fissures; hyperteloritic	flat occipitus; low-set ears	long face; flattened malar regions; short philtrum; everted lips; broad nasal root; small chin	fifth-finger clinodactyly	partial second- and third-toe syndactyly
MR43-VI:1	c.1418G>A; p.Trp473*	NK ^a	NK ^a	NK ^a	NK ^a	NK ^a
MR43-VI:5	c.1418G>A; p.Trp473*	down-slanting palpebral fissures; hyperteloritic	NK ^a	long face; flattened malar regions; short philtrum; broad nasal root; small chin	NK ^a	NK ^a
8600060-V:1	c.1000C>T; p.Arg334Cys	down-slanting palpebral fissures; hyperteloritic; broad, long eyebrows		flat philtrum; thin upper lip; triangular, pointed chin; prominent nose		
8600060-V:2	c.1000C>T; p.Arg334Cys	down-slanting palpebral fissures; hyperteloritic; broad, long eyebrows	mild dolicocephaly	flat philtrum; thin upper lip; triangular, pointed chin; prominent nose		
8600060-V:3	c.1000C>T; p.Arg334Cys	down-slanting palpebral fissures; hyperteloritic; broad, long eyebrows	mild dolicocephaly	flat philtrum; thin upper lip; triangular, pointed chin; prominent nose		

Nucleotide sequence uses the reference mRNA sequence for *MAN1B1*, NM_016219.3. Photographs will be made available to researchers by the corresponding author, upon signing a nondisclosure agreement.

^a NK: not known (photographs or clinical information is unavailable to researchers).

^b Now deceased (from hepatitis C).

cerebellar and cerebral sulci that are mildly prominent but not enough so to constitute volume loss or atrophy, especially given that the ventricles are normal in size (see [Figure S1](#)). MR8-IV:1 died in 2009 (from hepatitis C). MR8-IV:2 (age 10 years) has a slightly higher level of functioning than MR7-IV:5 and IV:6 and speaks in sentences. He is toilet trained can dress himself if supervised. He attends school, can read words of one or two syllables, and can count to ten. MR9-II:1 (age 18 years) has a developmental trajectory similar to that of MR7-IV:5 and IV:6. He is able to speak in sentences but is unable to read or write. Inappropriate sexualized behavior is a major cause for concern to the family. He has epileptic seizures, which started at 10 years of age. He has at least one episode per week. Episodes begin with twitching of the facial muscles, then become generalized, at which point he loses consciousness. The seizures have a tonic and then clonic phase and last for several

minutes, after which he takes 3 to 4 hr to recover. During this period he is drowsy and disorientated. He frequently becomes incontinent during the fit. Sometimes the family members are able to avert a fit by distracting his attention. MR9-II:3 (age 9 years) has a developmental trajectory similar to that of her sibling, MR9-II:1, but no epilepsy reported to date. Electroencephalographs were not available for MR9-II:1. Verbal and physical aggression is a common feature among affected individuals from MR7 and MR9, but not for individual MR8-IV:2. Mild dysmorphic features were noted for members of families MR7, MR8, and MR9 (see [Table 2](#)); however, these cannot be excluded as familial traits, and the families are categorized as nonsyndromic.

It was established that MR7, MR8, and MR9 all shared a HBD region on chromosome 9 (9q34.3; between SNPs rs11103117 and rs12238423 [nucleotides 137,667,734–139,811,104 (UCSC March 2006)]; [Figure 2](#)), and the locus

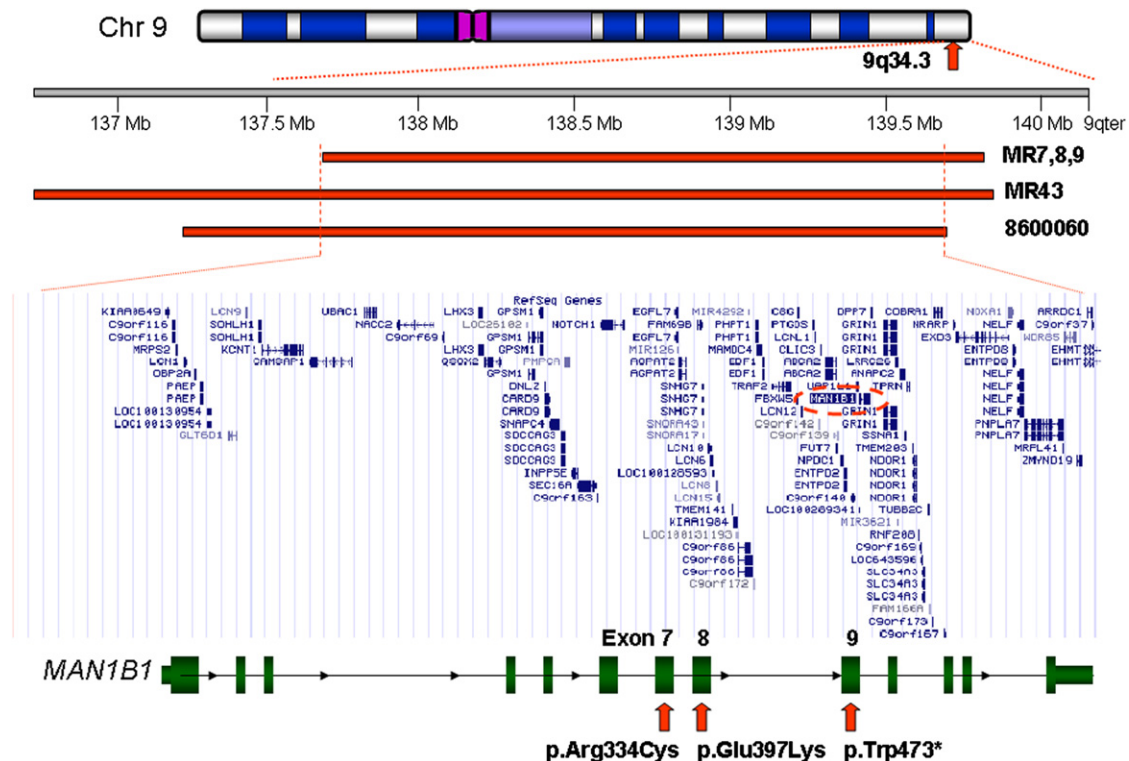


Figure 2. Ideogrammatic Representation Indicating the Overlapping HBD Regions for All Three Families on 9q34.3, the RefSeq Genes within the Common Region, the Genomic Organization of *MAN1B1*, and the Location within the Gene for the Three mutations

was assigned as MRT15.⁵ The critical region was 2.14 Mb, and linkage analysis gave LOD score = 4.8.⁵ Although we were unable to establish familial connections in the ancestry of families MR7, MR8, and MR9, comparison of haplotypes suggests a common founder.

MR43 has three affected members, VI:1, VI:4, and VI:5 (ages 16, 9, and 8, respectively) with milder, higher functioning intellectual disability. On assessment, VI:4 showed delayed developmental milestones; he sat at 9 months and walked at 2 years. He can talk in sentences and is independent in terms of self care. He can help with chores around the farm. He is sociable and caring but can become verbally and physically aggressive upon provocation and can damage property. On the basis of the assessment, he has mild ID (estimated IQ 55–65). He has epileptic seizures, which started at 6 years of age. These are primary generalized seizures, which start with eye rolling followed by loss of consciousness. Electroencephalographs were not available. His physical appearance shows facial dysmorphic features, including down-slanting palpebral fissures, hypertelorism, a long face, flattened malar regions, a short philtrum, everted lips, a broad nasal root, and a small chin (Table 2). He has fifth-finger clinodactyly. His elder sister (IV:1) and younger brother (IV:5) have a similar developmental trajectory and level of functioning and similar dysmorphic features (IV:1 was not assessed), but no history of epilepsy. Brain computed tomography (CT) scan for an VI:4 was examined by a radiologist (GI) and showed no apparent abnormalities, including no

obvious white-matter abnormalities or obvious cerebellar atrophy.

For MR43, an HBD region overlapped with that of MRT15 (9q34.3-qter: SNPs rs11103399–rs11137379; nucleotides 136,609,628–140,147,760; Figure 2).

The Iranian family, 8600060, had three affected members of ages 11, 23, and 25 years. They had mild to moderate NS-ARID, which was ascertained and mapped as previously described⁶ to 9q34.3 (SNPs rs2031825–rs11137163, 137,166,820–139,653,199; Figure 2), and linkage analysis gave an LOD score = 3.1.⁶ On assessment, the 11-year-old boy (V:1) showed only slight developmental delay. He started to walk at 1.5 years of age. His speech started at age 2.5. He speaks well and can count and use money. He can dress and wash independently and perform day-to-day tasks such as answering the telephone. He has no history of epilepsy, and just mild ID. His sister, V:3, a 23-year-old, has slight developmental delay, but it is more severe than that of her brother, and she shows aggressive behavior. She can dress and wash independently but cannot count or use money. Her speech is less advanced than her brother's. She has no history of epilepsy. Her older sister, V:2, age 25 years, was more delayed than her younger siblings. She began walking at 4 years. She speaks in sentences but cannot count or use money. She has problems with overeating and aggression and has a history of generalized epilepsy. Study of photographs of affected individuals from this family suggested common facial features, including a flat philtrum and

334		397
MAN1A1 FEVNI IR FGGLLSAYYLSGEEI 300		SS-ILAE FG TLHLEFHMHL 360
MAN1A2 FEVNI IR FIGGLLAAYYLSGEEI 286		SS-ILAE FG TLHMEFIHL 346
MAN1C1 FEVNI IR YIGGLLSAFYLTGEEV 279		SSSILAE FG SLHLEFLHL 337
MAN1B1 FESTI IR ILGGLLSAYHLSGDSL 350		TV-- AE VTSIQLEFREL 408

Figure 3. Paralogues of MAN1B1 in Humans Are Also Type 1 Mannosidases and Contain Glycosyl Hydrolase Family 47 Domains
Arg334 and Glu397 are both conserved across these paralogous sequences. Alignments were performed with CLUSTAL 2.1 multiple sequence alignment.

prominent nose. However, because photographs of other family members were not available, these cannot be excluded as familial traits, and the family is categorized as nonsyndromic (see [Tables 1 and 2](#)).

When all the Pakistani and Iranian families are taken into consideration, a combined critical region spanning nucleotides 137,667,734–139,653,199 (~2.0 Mb) was likely to harbor a gene in which mutations lead to NS-ARID (see [Figure 2](#)). This critical region contains 82 RefSeq coding genes (UCSC hg18 March 2006).

For the Iranian family, we performed next-generation sequencing by enrichment of the exonic regions within the linkage interval by using a custom Agilent SureSelect array, followed by sequencing with the Illumina Genome Analyzer II platform. Analysis of prospective changes from the critical region indicated DNA variants in just three genes: *GPSM1* [MIM 609491] (one base pair deleted at Chr9:139,235,486 (hg19); RefSeq number NM_015597.4; c.1243 delA [p. Thr415Glnfs*55]), *SOHLH1* [MIM 610224] (missense change: c. 916C>A [p.Leu306Met]; RefSeq number NM_001101677.1) and *MAN1B1* [MIM 604346] (missense change: c.1000C>T [p.Arg334Cys]; RefSeq number NM_016219.3). However, only the *MAN1B1* change segregated in pedigree 8600060. This change, which occurs in exon 7, was not present in 155 Iranians or 191 Germans.

For the Pakistani families, a number of genes were selected as candidates for sequencing, but sequence analysis excluded *INPP5E* [MIM 613037], *GRIN1* [MIM 138249], *ABCA2* [MIM 600047], *CACN1A8*, *EHTM1*, *KCNT1* [MIM 608167], *CAMSAP1*, *OLFM1* [MIM 605336], *GLT6D1* [MIM 613699], *SEC16A* [MIM 612854], *LHX3* [MIM 600577], *ANAPC2* [MIM 606946], *PNPLA7* [MIM 612122], and *TUBB2C* [MIM 602660]. Sequence analysis of *GPSM1*, *SOHLH1*, and *MAN1B1* was then undertaken. The same 1 bp deletion was observed in *GPSM1* in an affected member of MR43, however this did not segregate, and it was present in homozygous form in an unaffected sibling. This deletion occurs within the last exon of *GPSM1*, but the location is only present within the coding region for one of the four known isoforms of the gene (RefSeq; UCSC Hg18, March 2006) and is either within intron 9 or upstream for the other isoforms. No changes were found in *SOHLH1*, but a missense mutation in exon 8 and a nonsense mutation in exon 9 were found in *MAN1B1* for MR7, MR8, MR9 (NM_016219.3: c. 1189G>A [p.Glu397Lys]), and MR43 (NM_016219.3: c.1418G>A [p.Trp473*]), respectively.

Both the Arg334 and Glu397 residues are conserved across evolution of the animal kingdom ([Figure 1](#)). Furthermore, these residues are also conserved across paralogous

sequences that are known to have similar enzymatic functions but in different cellular contexts ([Figure 3](#)). The c.1189G>A substitution creates a StyI restriction endonuclease cutting site, and the c.1418G>A destroys a BamHI restriction site. We used PCR followed by either StyI or BamHI digestion to screen a cohort of 252 Pakistani controls, but the mutant alleles were not detected. None of the three *MAN1B1* substitutions have been reported in any SNP databases to date and were not present in the 1000 Genomes Version 60.37e.

MAN1B1 encodes the protein endoplasmic reticulum mannosyl-oligosaccharide 1,2- α -mannosidase (ERManI; EC = 3.2.1.113). ERManI and other class 1 α -mannosidases are members of the glycosyl hydrolase family 47 (GH47), are believed to be key enzymes involved in the maturation of N-glycans in the secretory pathway,^{7,8} and contribute to the timing and disposal of misfolded glycoproteins through the endoplasmic-reticulum-associated degradation pathway.⁹ Enzyme activity involves a direct Ca²⁺-mediated interaction between the enzyme and substrate,⁸ and the protein forms an (α)₇ barrel structure with the Ca²⁺ ion bound at the core of an active-site pocket in the center of the barrel.⁹ The Arg334 residue, which is mutated in the Iranian family, is believed to be located at the base of the active-site pocket.⁹ The Glu397 residue, which is mutated in the Pakistani MR7, MR8, and MR9 families, is also located at this active-site pocket, where it interacts with glycan substrates (see [Figure 4](#); also see [Movie S1](#)).⁹ Arg334 interacts via hydrogen bonding with the 4' OH on the glycone in the –1 subsite, and the Glu397 residue is involved in hydrogen bonding to the 6' OH of the +1 mannose residue. Each of these interactions probably makes a minor energetic contribution relative to others in the active site,¹⁰ but the p.Arg334Cys mutation might alter the electrostatics in the active site in ways that are not easily predicted. The p.Glu397Lys mutation is likely to be disrupting, partially because of the loss of hydrogen bonding to the side chain but also because the longer Lys side chain would probably provide steric hindrance to docking of the +1 residue. Thus, both missense mutations would probably interfere indirectly with the enzyme's ability to bind and recognize the appropriate oligosaccharide substrate. The cDNA encoding the *MAN1B1* catalytic domain was used for site-directed mutagenesis, as described previously.⁹ The constructs were used for transient transfection of HEK293 cells. The wild-type (wt) construct was secreted well and was purified on a Ni²⁺-NTA column; however, the Cys334 construct was expressed and secreted at ~20% of that of wt levels, and Lys397 was expressed and secreted at <5% of wt levels. Although the amount of expressed

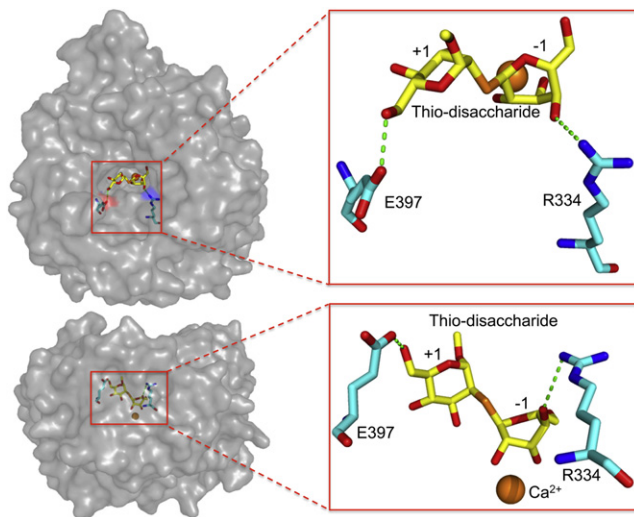


Figure 4. 3D Protein Structure for MAN1B1 Indicating the Location of Missense Residues

The structure shows a transparent surface representation of the enzyme in gray and a stick representation of the glycan substrate (thiodisaccharide) in yellow; the mannose residues in the -1 and $+1$ subsites are indicated. The Ca^{2+} ion interacting with the $2'$ OH and $3'$ OH of the glycone residue in the -1 subsite is indicated with orange spacefill. The two amino acids that are mutated, Arg334 and Glu397, are shown in stick form in blue and are clearly located close together; they are believed to form hydrogen bonds with the substrate. The molecular display was generated with MacPymol 1.3 and the PDB file 1X9D.⁹ See [Movie S1](#) for a video of the molecular structure.

protein for the Lys397 mutant was too low for enzyme activity to be measured, kinetic analysis of the Cys334 mutant indicated a 1327-fold reduction in k_{cat} , a 6.4-fold reduction in K_m , and a 205-fold decrease in K_{cat}/K_m (Table 3). It is highly probable that these missense mutations destabilize the protein and thus negatively affect folding and secretion.

The Trp473 nonsense mutation identified in family MR43 is thus likely to result in a truncated form of the protein missing many of the residues required for substrate recognition, as well as the Ca^{2+} -binding Thr688 residue. Furthermore, it occurs in exon 9 of a 13 exon gene and therefore most likely leads to nonsense-mediated RNA decay so that only a little mRNA, if any, would survive to protein translation. However, mRNA was not available from any MR43 family members, and as such, we were unable to confirm this experimentally.

The lysosomal-storage disease mannosidosis (MIM 248500 for α mannosidosis; MIM 245100 for β mannosidosis) is an autosomal-recessive condition caused by mutations within genes for lysosomal α -mannosidase enzyme *MAN2B1* (MIM 609458) and lysosomal β -mannosidase enzyme *MANBA* (MIM 609489). Alpha-mannosidosis is a rare condition, with an estimated prevalence between 1:500,000 and 1:750,000,^{11,12} and it varies in severity from prenatal lethal forms to late-onset forms with slow progression. The main features are intellectual disability,

Table 3. Kinetic Analysis of the MAN1B1 Cys334 Mutation

Enzyme Form	k_{cat} (s^{-1})	K_m (μM)	k_{cat}/K_m (s^{-1}/M)	k_{cat}/K_m ($k_{\text{cat}}/K_m(\text{wt})$)
Wild-type	1.31 ± 0.24	133 ± 4	9850	1
Cys334	0.000987 ± 0.000015	20.6 ± 1.3	47.9	0.00486

The coding region for the catalytic domain of MAN1B1⁹ was subcloned into a modified version of the pXLG vector¹⁵ encoding an NH_2 -terminal signal sequence from the *T. cruzi* lysosomal mannosidase,¹⁶ 8 $\times$ His tag, Strep II tag, and then a TEV protease cleavage site and the *MAN1B1* coding region. The construct was transiently transfected into HEK293 suspension culture cells (Freestyle 293-F, Invitrogen) via the TransIT Pro transfection reagent (Mirus). Five days after transfection the conditioned medium was harvested, and the recombinant enzyme was purified by Ni^{2+} -NTA chromatography (QIAGEN) and concentrated by ultrafiltration. Enzyme assays were performed with Man₉GlcNAc₂-PA as a substrate.^{9,10} Substrate concentrations varying from 5 μM to 1 mM. Enzymatic cleavage was monitored by HPLC, and kinetic values were determined with SigmaPlot.

hearing loss, myopathy, skeletal abnormalities, and facial dysmorphic features, as well as vulnerability to infections (see GeneReviews in [Web Resources](#)). Brain magnetic resonance imaging (MRI) of α -mannosidosis patients typically show white-matter abnormalities as well as cerebellar atrophy and other structural anomalies (see GeneReviews in [Web Resources](#)). β -mannosidosis is even rarer, and clinical features include intellectual disability, behavioral abnormalities, and hearing loss; occasionally, there is also facial dysmorphism, skeletal deformation, susceptibility to respiratory infections, and skin lesions.¹⁴ The clinical features of the four families identified with *MAN1B1* missense mutations indicate nonsyndromic ARID because only mild dysmorphic features that could not be excluded as family traits were observed, whereas clear dysmorphic features were noted for the family with the nonsense mutation (MR43; see Table 2; photos of patients from MR7 (IV:5, IV:6), MR8 (IV:2), MR9 (II:1, II:3), and MR43 (VI:4, VI:5), as well as individuals V-1, V:2, and V:3 from family 8600060, were examined by R.W.). A more detailed physical examination might be required in order to reveal more subtle anomalies.

Thus, we believe that mutation of this endoplasmic reticulum mannosidase gene results in NS-ARID rather than mannosidosis, and thus the disease mechanism is different from disease-causing mutations in the lysosomal α and β mannosidase genes *MAN2B1* and *MANBA*. The identification of mutations causing NS-ARID in *MAN1B1* suggests that the ER-associated degradation (ERAD) pathway is a new disease-associated pathway, and this leads us to further speculate that disruption of other ERAD pathway molecules might result in similar clinical presentation.

Thus, mutations in *MAN1B1* have been found to cause NS-ARID. Furthermore, this is one of the rare cases of NS-ARID genes with increased mutation frequency, and it is tempting to speculate that future investigations into the molecular causes of NS-ARID in larger numbers of patients might demonstrate that disruption of *MAN1B1* accounts for more than one percent of the patients.

Supplemental Data

Supplemental Data include one figure and one movie and can be found with this article online at <http://www.cell.com/AJHG/>.

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Web Resources

The URLs for data presented herein are as follows:

GeneReviews, <http://www.ncbi.nlm.nih.gov/sites/GeneTests/review?db=GeneTests>

Online Mendelian Inheritance in Man (OMIM), <http://www.OMIM.org>

UCSC Genome Bioinformatics, <http://genome.ucsc.edu>

1000 Genomes, <http://www.1000genomes.org>

CLUSTAL 2.1 multiple sequence alignment, <http://www.ebi.ac.uk>

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Update

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Mutations in the Alpha 1,2-Mannosidase Gene, *MAN1B1*, Cause Autosomal-Recessive Intellectual Disability

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The affiliation for Muhammad Ansar was incorrectly listed in the original paper. It should have appeared as Department of Biochemistry, Quaid-I-Azam University, Islamabad, Pakistan. The authors regret the error.

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